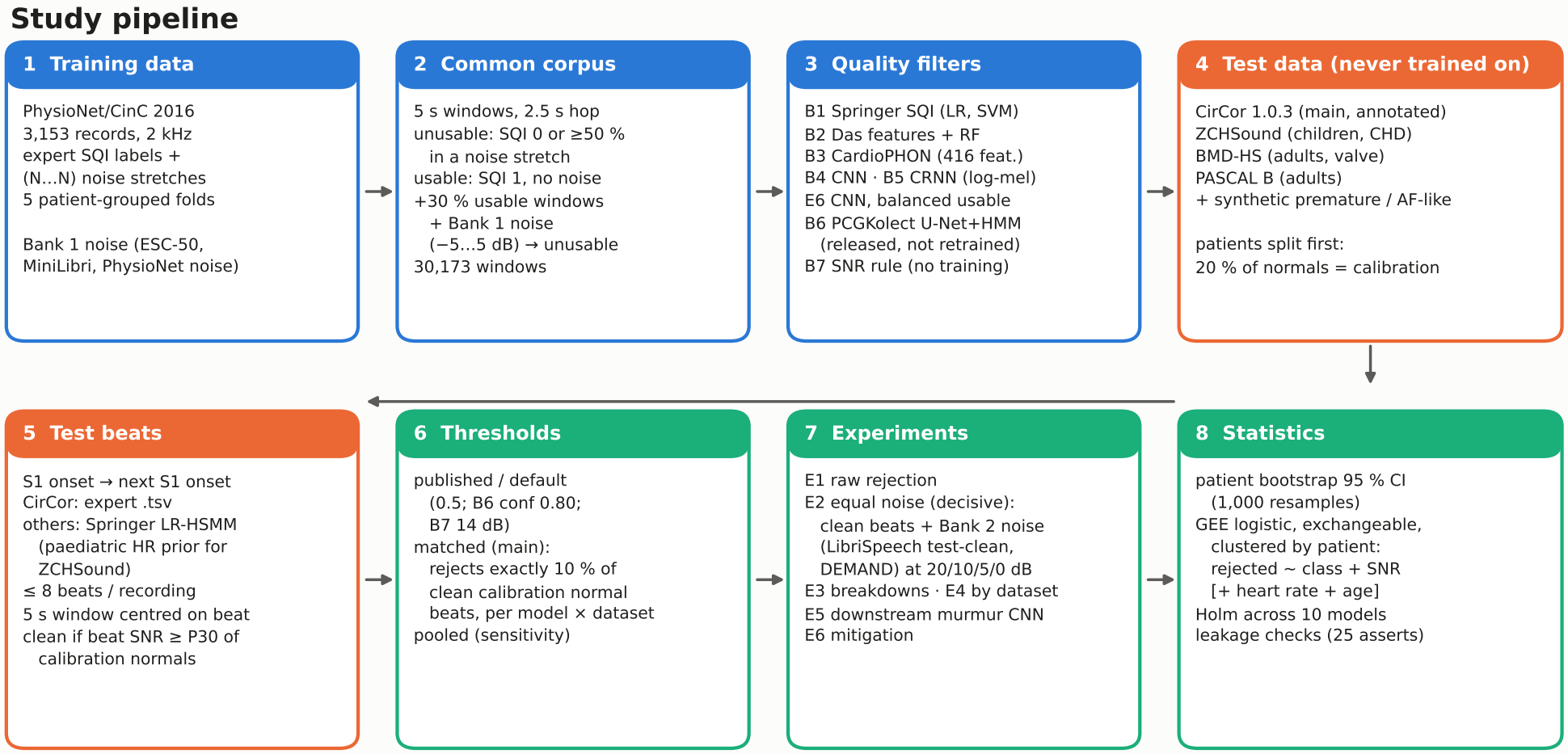
Study pipeline
1 Training data
PhysioNet/CinC 2016
3,153 records, 2 kHz
expert SQI labels +
(N…N) noise stretches
5 patient-grouped folds
Bank 1 noise (ESC-50,
MiniLibri, PhysioNet noise)
2 Common corpus
5 s windows, 2.5 s hop
unusable: SQI 0 or ≥50 %
in a noise stretch
usable: SQI 1, no noise
+30 % usable windows
+ Bank 1 noise
(−5…5 dB) → unusable
30,173 windows
3 Quality filters
B1 Springer SQI (LR, SVM)
B2 Das features + RF
B3 CardioPHON (416 feat.)
B4 CNN · B5 CRNN (log-mel)
E6 CNN, balanced usable
B6 PCGKolect U-Net+HMM
(released, not retrained)
B7 SNR rule (no training)
4 Test data (never trained on)
CirCor 1.0.3 (main, annotated)
ZCHSound (children, CHD)
BMD-HS (adults, valve)
PASCAL B (adults)
+ synthetic premature / AF-like
patients split first:
20 % of normals = calibration
5 Test beats
S1 onset → next S1 onset
CirCor: expert .tsv
others: Springer LR-HSMM
(paediatric HR prior for
ZCHSound)
≤ 8 beats / recording
5 s window centred on beat
clean if beat SNR ≥ P30 of
calibration normals
6 Thresholds
published / default
(0.5; B6 conf 0.80;
B7 14 dB)
matched (main):
rejects exactly 10 % of
clean calibration normal
beats, per model × dataset
pooled (sensitivity)
7 Experiments
E1 raw rejection
E2 equal noise (decisive):
clean beats + Bank 2 noise
(LibriSpeech test-clean,
DEMAND) at 20/10/5/0 dB
E3 breakdowns · E4 by dataset
E5 downstream murmur CNN
E6 mitigation
8 Statistics
patient bootstrap 95 % CI
(1,000 resamples)
GEE logistic, exchangeable,
clustered by patient:
rejected ~ class + SNR
[+ heart rate + age]
Holm across 10 models
leakage checks (25 asserts)